APPENDIX I. REPUBLIC OF YEMEN: DEBT SUSTAINABILITY—UPDATE

1. **In view of Yemen's request for access to Fund resources under the Extended Credit Facility, Fund staff has updated the assessment of Yemen's debt sustainability**. The last joint Bank-Fund debt sustainability analysis found that, although Yemen's debt ratios were below most of the indicative policy-dependent thresholds, dwindling oil production presented significant risks over the medium and long term as the present value of debt-to-exports threshold was breached over the long term. This assessment builds on the previous exercise. It incorporates the previous projections of debt service on the stock of outstanding external and domestic debt and includes debt service projections on new domestic and external debt obligations, including the proposed use of Fund resources. The macroeconomic assumptions underlying the assessment have also been revised in line with program parameters.

2. **Yemen's risk of debt distress is broadly unchanged relative to last year's assessment**. Yemen's external debt and debt service are projected to slightly increase as a share of GDP, exports, and revenues during 2010 and beyond, mostly because of lower export receipts. The use of Fund resources will have only a modest impact on these ratios. With the proposed use of Fund resource, the debt indicators remain below their thresholds, except the present value of debt-to-exports ratio in the baseline scenario.

Table 1.Yemen: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	40.4	36.4	51.0			46.3	46.3	46.0	45.2	45.0	45.7		39.3	48.8	
o/w foreign-currency denominated	26.9	21.9	24.2			21.3	21.2	21.4	21.4	22.1	22.9		24.0	20.9	
Change in public sector debt	-0.4	-4.0	14.6			-4.7	0.0	-0.3	-0.8	-0.2	0.7		-1.8	-1.6	
Identified debt-creating flows	2.1	-3.4	13.0			-4.8	0.1	-0.4	-0.7	-0.3	0.6		-1.6	-1.3	
Primary deficit	4.9	2.1	7.7	-0.2	4.8	3.3	2.6	2.3	0.9	1.0	1.8	2.0	-0.4	0.6	1.5
Revenue and grants	33.2	36.7	25.0			26.8	25.2	23.9	23.2	22.6	21.8		23.0	18.9	
of which: grants	0.3	0.3	0.4			1.2	1.5	1.5	1.5	1.4	1.3		0.5	0.5	
Primary (noninterest) expenditure	38.0	38.8	32.7			30.1	27.8	26.1	24.2	23.7	23.6		22.6	19.5	
Automatic debt dynamics	-2.7	-5.6	5.3			-8.1	-2.5	-2.7	-1.7	-1.3	-1.2		-1.2	-1.8	
Contribution from interest rate/growth differential	-0.8	-1.7	2.1			-6.3	-2.0	-2.1	-1.5	-1.5	-1.5		-1.7	-1.8	
of which: contribution from average real interest rate	0.5	-0.2	3.5			-2.5	-0.2	-0.3	0.4	0.4	0.5		0.2	0.4	
of which: contribution from real GDP growth	-1.3	-1.4	-1.4			-3.8	-1.8	-1.8	-1.9	-1.9	-2.0		-1.8	-2.2	
Contribution from real exchange rate depreciation	-1.9	-3.9	3.2			-1.8	-0.5	-0.6	-0.2	0.2	0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.5	-0.6	1.5			0.1	0.0	0.1	-0.1	0.1	0.1		-0.2	-0.4	
Other Sustainability Indicators															
PV of public sector debt	13.5	14.5	42.5			40.0	40.4	40.2	39.7	39.2	39.6		32.8	43.7	
o/w foreign-currency denominated	0.0	0.0	17.2			15.4	15.8	16.1	16.3	16.8	17.4		17.5	15.8	
o/w external	...	...	17.2			15.4	15.8	16.1	16.3	16.8	17.4		17.5	15.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.1	5.3	10.9			6.2	5.6	5.3	4.1	3.8	4.5		1.9	3.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	40.6	39.5	169.9			149.4	160.2	168.5	170.8	173.2	182.2		142.7	230.9	
PV of public sector debt-to-revenue ratio (in percent)	41.0	39.7	172.7			156.5	170.3	179.5	182.6	185.0	194.1		145.6	236.5	
o/w external 3/	…	…	69.8			60.4	66.5	72.0	75.2	79.3	85.3		77.7	85.5	
Debt service-to-revenue and grants ratio (in percent) 4/	9.9	8.6	12.9			11.0	12.1	12.6	13.5	12.2	12.3		10.2	14.8	
Debt service-to-revenue ratio (in percent) 4/	10.0	8.7	13.1			11.5	12.8	13.4	14.5	13.0	13.1		10.4	15.2	
Primary deficit that stabilizes the debt-to-GDP ratio	5.3	6.1	-6.9			8.0	2.5	2.6	1.7	1.2	1.1		1.4	2.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.3	3.6	3.9	4.1	1.0	8.0	4.1	4.1	4.4	4.5	4.5	4.9	4.7	4.5	4.4
Average nominal interest rate on forex debt (in percent)	1.2	1.2	1.1	1.2	0.3	1.3	1.2	1.2	1.2	1.2	1.1	1.2	1.0	6.5	1.4
Average real interest rate on domestic debt (in percent)	7.7	-0.5	25.6	5.4	9.4	-9.4	0.1	-0.4	2.5	2.7	3.1	-0.2	2.6	-1.8	2.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.9	-15.1	15.2	-5.0	9.2	-8.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	10.9	20.4	-8.7	11.4	9.3	22.4	9.9	10.6	8.1	6.2	5.7	10.5	6.0	4.4	4.8
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	34.9	34.7	34.9	34.8	35.8	35.8	35.1	35.8	35.8	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

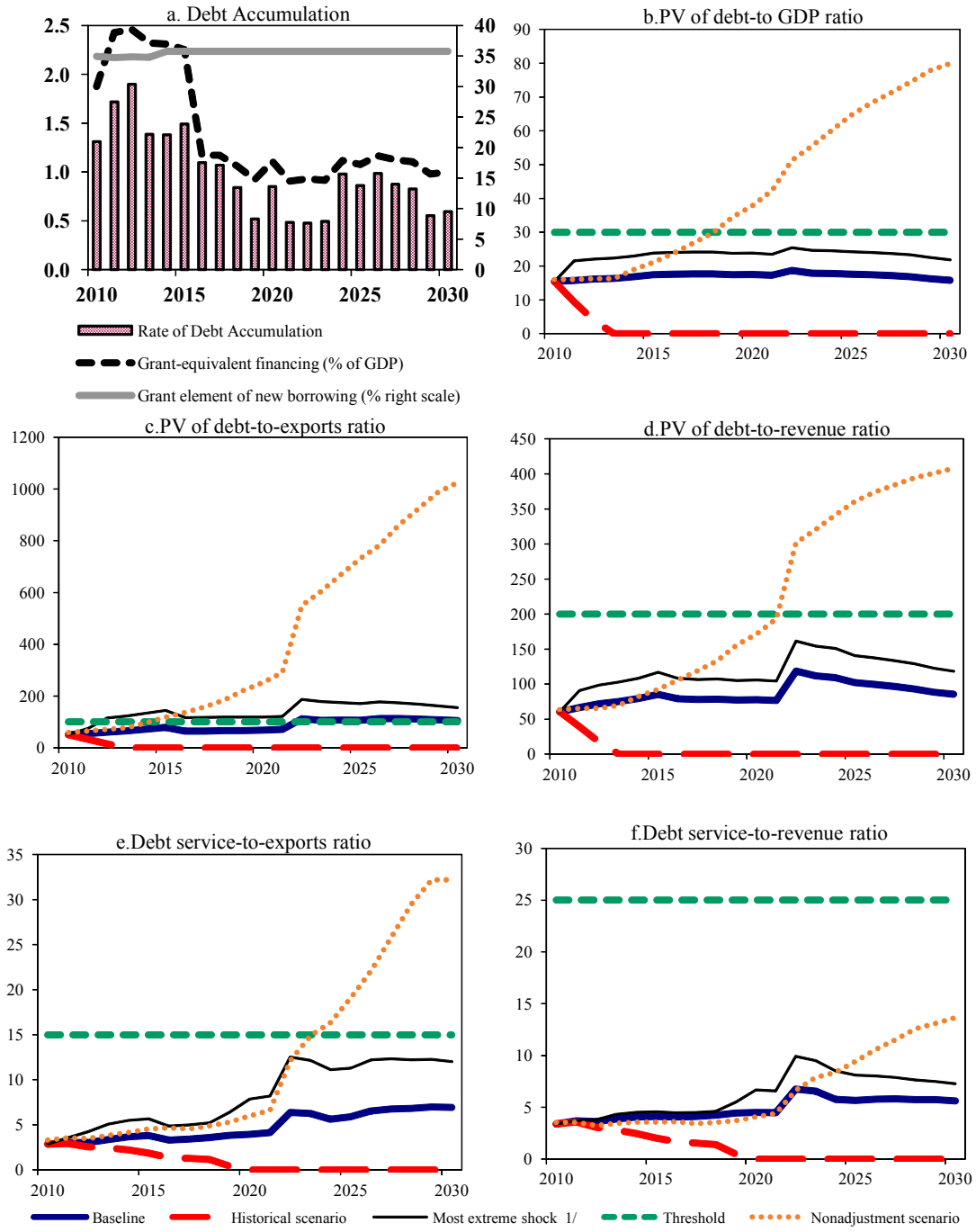
Table 2.: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average 0	Standard Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**26.9**	**21.9**	**24.2**			**21.7**	**23.4**	**23.3**	**23.7**	**24.8**	**26.2**		**25.6**	**23.8**	
o/w public and publicly guaranteed (PPG)	26.9	21.9	24.2			21.3	21.2	21.4	21.4	22.1	22.9		24.0	20.9	
Change in external debt	0.0	-5.0	2.2			-2.4	1.6	0.0	0.3	1.1	1.4		-0.4	-0.9	
Identified net debt-creating flows	-1.6	-2.4	12.9			6.0	5.1	5.5	4.8	4.2	4.4		0.4	-0.2	
Non-interest current account deficit	**6.7**	**4.4**	**10.4**	**-1.5**	**7.2**	**4.9**	**4.5**	**4.3**	**4.3**	**4.5**	**4.6**		**1.6**	**2.5**	4.4
Deficit in balance of goods and services	7.3	5.6	12.1			4.7	4.5	5.4	6.6	7.9	9.3		4.0	10.0	
Exports	35.9	37.8	28.2			30.1	29.0	26.8	24.9	23.3	22.1		25.7	15.0	
Imports	43.2	43.4	40.3			34.8	33.4	32.2	31.5	31.2	31.3		29.6	24.9	
Net current transfers (negative = inflow)	-6.6	-8.0	-6.0	-9.8	3.0	-5.9	-5.7	-6.0	-6.7	-7.4	-7.6		-6.7	-9.1	-7.9
o/w official	-0.7	-3.0	-1.6			-1.7	-1.3	-1.5	-1.6	-1.5	-1.5		-0.5	-0.5	
Other current account flows (negative = net inflow)	5.9	6.9	4.4			6.1	5.7	4.9	4.4	4.1	3.0		4.3	1.6	
Net FDI (negative = inflow)	**-5.3**	**-1.7**	**0.7**	**-1.4**	**2.5**	**2.4**	**1.2**	**1.8**	**1.2**	**0.4**	**0.6**		**-0.2**	**-1.8**	-1.3
Endogenous debt dynamics 2/	**-2.9**	**-5.0**	**1.8**			**-1.4**	**-0.6**	**-0.6**	**-0.7**	**-0.8**	**-0.8**		**-0.9**	**-0.8**	
Contribution from nominal interest rate	0.3	0.3	0.3			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.2	
Contribution from real GDP growth	-0.8	-0.8	-0.9			-1.6	-0.8	-0.9	-0.9	-1.0	-1.1		-1.2	-1.0	
Contribution from price and exchange rate changes	-2.4	-4.5	2.5			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**1.5**	**-2.7**	**-10.7**			**-8.4**	**-3.5**	**-5.5**	**-4.5**	**-3.1**	**-3.0**		**-0.9**	**-0.7**	
o/w exceptional financing	0.0	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	18.4			15.8	17.9	18.1	18.7	19.5	20.7		19.1	18.7	
In percent of exports	...	...	65.1			52.6	61.7	67.5	75.0	83.5	93.7		74.5	125.2	
PV of PPG external debt	**...**	**...**	**17.2**			**15.4**	**15.8**	**16.1**	**16.3**	**16.8**	**17.4**		**17.5**	**15.8**	
In percent of exports	**...**	**...**	**60.8**			**51.2**	**54.4**	**60.3**	**65.7**	**72.1**	**79.0**		**68.3**	**105.4**	
In percent of government revenues	**...**	**...**	**69.8**			**60.4**	**66.5**	**72.0**	**75.2**	**79.3**	**85.3**		**77.7**	**85.5**	
Debt service-to-exports ratio (in percent)	**3.5**	**2.7**	**3.5**			**2.9**	**3.0**	**3.0**	**3.4**	**3.7**	**3.8**		**3.9**	**6.9**	
PPG debt service-to-exports ratio (in percent)	**3.5**	**2.7**	**3.5**			**2.9**	**3.0**	**3.0**	**3.4**	**3.7**	**3.8**		**3.9**	**6.9**	
PPG debt service-to-revenue ratio (in percent)	**3.8**	**2.8**	**4.1**			**3.4**	**3.7**	**3.6**	**3.9**	**4.1**	**4.1**		**4.5**	**5.6**	
Total gross financing need (Billions of U.S. dollars)	0.6	1.0	3.0			2.5	2.2	2.5	2.5	2.4	2.6		1.3	1.5	
Non-interest current account deficit that stabilizes debt ratio	6.7	9.4	8.2			7.4	2.9	4.3	4.0	3.4	3.2		2.0	3.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.3	3.6	3.9	4.1	1.0	8.0	4.1	4.1	4.4	4.5	4.5	4.9	4.7	4.5	4.4
GDP deflator in US dollar terms (change in percent)	9.9	19.9	-10.1	8.6	9.3	11.1	5.0	5.4	3.0	1.2	0.7	4.4	0.0	2.4	0.7
Effective interest rate (percent) 5/	1.3	1.3	1.2	1.2	0.3	1.2	1.2	1.1	1.1	1.1	1.0	1.1	1.0	0.9	0.9
Growth of exports of G&S (US dollar terms, in percent)	-1.2	31.0	-30.3	12.7	23.5	27.9	5.2	1.4	-0.2	-1.0	-0.5	5.5	2.7	6.2	3.2
Growth of imports of G&S (US dollar terms, in percent)	20.3	24.8	-13.3	14.2	12.2	3.8	4.8	5.5	5.3	4.5	5.6	5.0	3.7	3.0	3.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	34.9	34.7	34.9	34.8	35.8	35.8	35.1	35.8	35.8	35.8
Government revenues (excluding grants, in percent of GDP)	32.8	36.5	24.6			25.5	23.7	22.4	21.7	21.2	20.4		22.5	18.5	19.4
Aid flows (in Billions of US dollars) 7/	0.4	0.3	0.5			0.9	1.3	1.5	1.4	1.6	1.7		1.3	1.8	
o/w Grants	0.1	0.1	0.1			0.4	0.5	0.5	0.6	0.6	0.6		0.2	0.4	
o/w Concessional loans	0.4	0.3	0.4			0.5	0.8	0.9	0.8	1.0	1.1		1.0	1.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.9	2.4	2.5	2.3	2.3	2.3		1.1	1.0	1.1
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			60.3	58.0	57.0	60.0	59.6	58.0		48.4	50.3	49.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	21.7	26.9	25.1			30.2	33.0	36.2	38.9	41.1	43.2		55.4	90.7	
Nominal dollar GDP growth	13.6	24.3	-6.6			20.0	9.3	9.6	7.5	5.7	5.3	9.6	4.6	7.0	5.1
PV of PPG external debt (in Billions of US dollars)			4.2			4.6	5.1	5.7	6.2	6.7	7.3		9.4	14.2	
(PVt-PVt-1)/GDPt-1 (in percent)						1.3	1.7	1.9	1.4	1.4	1.5	1.5	0.9	0.6	0.8
Gross workers' remittances (Billions of US dollars)	1.3	1.4	1.2			1.3	1.5	1.7	2.1	2.5	2.8		3.6	8.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	16.4			14.8	15.1	15.4	15.5	15.8	16.4		16.5	14.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	52.1			44.6	46.9	51.2	54.0	57.2	61.3		54.6	65.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.0			2.5	2.6	2.6	2.8	2.9	3.0		3.2	4.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 1. Yemen: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Nonadjustment scenario shock; in c. to a Nonadjustment scenario shock; in d. to a Nonadjustment scenario shock; in e. to a Nonadjustment scenario shock and in figure f. to a Nonadjustment scenario shock